

WORK EXPERIENCE

AI/ML Computational Science Specialist, Accenture	09/2026 – Present
• Privacy-Safe Analytics Agent: Architected an end-to-end analytics platform with ingestion, PII/profanity redaction, hybrid (BM25 + Vector) Azure AI Search across 100K+ documents, multi-agent reasoning layer, shipped via Docker, GitHub Actions CI/CD with OpenTelemetry tracing; cutting retrieval latency ~75%, analyst effort 60–70%.	
AI/ML Computational Science Sr Analyst, Accenture	09/2024 – 08/2026
• Enterprise Data Q&A Agent: Engineered a text-to-SQL agent chain (Microsoft Copilot Studio → AI Foundry) that translates natural-language queries into T-SQL executed directly against Lakehouse SQL endpoints, enabling analytics on 10M+ records in <30s.	
• GenAI Business Doc Generator: Designed and delivered a multi-agent RAG system with LangGraph + LoRA fine-tuned LLM for business doc generation; with LangSmith tracing, ROUGE evals; cutting drafting time 60–70%	
Software Developer, Expro	09/2022 – 09/2024
• Well Performance Analytics Platform: Implemented a PyTorch LSTM-autoencoder pipeline for anomaly detection on well-sensor time-series data, improving risk detection by ~20% and reducing analysis time by ~40%.	
• Oil & Gas Insights Web Application: Designed a PySpark KPI pipeline on Azure Databricks using PCA, K-means, and Random Forest to derive KPI features from equipment telemetry, reducing manual reporting by ~50%.	
Data Scientist, Ganit Labs	03/2021 – 09/2022
• Equipment Failure Prediction Model: Developed predictive maintenance models on GCP for oilfield equipment with ~85% recall and ~30% less downtime.	
Biomedical Engineer, Portea, AOV, Care Group	08/2018 – 09/2020
• Led cross-functional teams in medical equipment maintenance and training programs.	

EDUCATION

MTech (Master of Technology), University of Calcutta	2016 – 2018
BTech (Bachelor of Technology), WBUT	2011 – 2015

TECHNICAL SKILLS

Languages: Python, SQL	Generative AI & Frameworks: Multi-agent orchestration, Retrieval-Augmented Generation (RAG), Prompt engineering, LLM fine-tuning, Observability, Evaluation, LangChain, LangGraph, CrewAI, Claude Agent SDK, MCP
AI Architecture & Governance: Solution architecture, multi-agent & RAG design patterns, AI guardrails, human-in-the-loop governance, cost/latency trade-off analysis	Machine Learning & Deep Learning: LSTM autoencoders, Isolation Forest, PCA, K-means, HDBSCAN, Random Forest, Decision Tree, Linear Regression, Logistic Regression, NLP
Cloud Platforms: Azure (AI Foundry, Storage, Fabric, Cosmos DB, AI Search, OpenAI, App Service, Function App, Logic Apps); GCP (Cloud Storage, BigQuery, App Engine)	Data & Retrieval: Vector search: FAISS, Pinecone, Azure AI Search · Graph: Neo4j · Platforms: Microsoft Fabric, Azure SQL, Azure Databricks
MLOps & DevOps: Docker, Git, GitHub Actions CI/CD, Azure DevOps, MLOps & LLMops pipelines, LangSmith, OpenTelemetry, OpenEvals, RAGAS, REST APIs	Frameworks & Libraries: PyTorch, scikit-learn, Hugging Face Transformers, Azure OpenAI / OpenAI SDK, FastAPI, Flask

CERTIFICATIONS & TRAININGS

• Claude Certified Architect – Foundations	• AI-102: Microsoft Azure AI Engineer Associate	• CS50AI: Introduction to Artificial Intelligence with Python — Harvard University
• AZ-900: Microsoft Azure Fundamentals	• AI-900: Microsoft Azure AI Fundamentals	• LLMOps, MLOps & CI/CD/CT Pipelines (Udemy)

PROJECTS & PUBLICATIONS

rti-engine – AI Pay Transparency Compliance Engine , https://github.com/trayan4/rti-engine Architected an 11-node LangGraph multi-agent system for pay-transparency compliance — 6 model-driven, 5 deliberately deterministic where no judgment is required — with tiered autonomy, human approval gates, cross-vendor review, figure-validation. MCP-exposed Neo4j graph; deployed on Azure, with Docker/GitHub Actions CI-CD, LangSmith/OTel tracing, openevals harness.
Autonomous Data Analyst , https://github.com/trayan4/autonomous-data-analyst Engineered a LangGraph multi-agent text-to-SQL system on Azure with governance-first routing, execution-repair validation, forced RCA decomposition, model tiering, and OpenTelemetry cost tracing; 87% on a 110-question golden set.
Multimodal Agentic Reasoning Assistant (MARA) , https://github.com/trayan4/MARA-AI-Agent Designed a multi-agent system coordinating RAG, Vision, Data & Web Search agents (3–5 parallel agent calls per query); hybrid FAISS+BM25 retrieval (recall@10 ~0.75), adaptive planning, FastAPI (p95 ~30s).
GPT-Style Language Model from Scratch , https://github.com/trayan4/llm-from-scratch Implemented a decoder-only 124M-parameter LLM from scratch in PyTorch — BPE tokenization, multi-head causal self-attention, training loop, instruction fine-tuning; trained on 124M tokens.

AWARDS

Spotlight Award, Client — delivery of the privacy-safe analytics agent platform	02/2026
Growth Catalyst Award, Accenture — technical proficiency and project delivery	12/2025